64-bit Word Memory Interface

word_data_in
[63 : 0]

write
(1 bit)

word_addr
[4 : 0]

64

5

write_word

64

1-to-25 Write Decoder / Demux

Selects Lane $A[x, y]$ where word_index $= 5y + x$

*Synchronous write gating (*write $= 1$*)*

Drives individual lane clock-enables and data bus

5

sel

25-to-1 Read Multiplexer

word_data_out $=$ state_words[word_addr]

(Zero-latency asynchronous multiplexer)

64

word_data_out [63:0]

Valid during S_IDLE / S_DONE

1600-bit Iterative Round Datapath

sel $=$ (state $=$ $=$ S_RUN)

0

State Input Multiplexer (2-to-1, 1600-bit)

1

Port 0: Word Write Insertion | Port 1: Round Feedback (S_RUN)

1600

1600-bit State Register File (25 × 64-bit D-FF Lanes)

State Array: $A[x, y]$ for $x, y \in \{0 \dots 4\}$ | Word Index: $i = 5y + x$

Synchronous active-low reset • Clock-gated lane / round enables

1600

1600

state_words

Step θ (Theta) — Column Parity Mixing & Diffusion

$C[x] = \bigoplus_{y=0}^{4} A[x, y], \quad D[x] = C[x-1] \oplus \mathrm{ROT}(C[x+1], 1), \quad A'[x, y] = A[x, y] \oplus D[x]$

Linear diffusion layer: 320 parallel 5-input XOR trees (3200 XOR gates total)

1600

Steps ρ & π (Rho & Pi) — Bit Rotation & Coordinate Permutation

$B[y, (2x + 3y) \bmod 5] = \mathrm{ROT}(A'[x, y],\ r[x, y])$

[Hardwired Wire Interconnect: 0 Logic Gates, 0 Silicon Area, 0 ns Delay]

1600

Step χ (Chi) — 5-bit Non-Linear Algebraic S-Box Layer

$A_\chi[x, y] = B[x, y] \oplus ((\neg B[x+1, y]) \cdot B[x+2, y])$

Non-linear algebraic degree 2: 320 parallel 5-bit S-boxes (1600 AOI/OAI/NAND gates)

1600

Step ι (Iota) — Round Constant Symmetry Breaking

$A_{\text{out}}[0, 0] = A_\chi[0, 0] \oplus RC[\text{round_cnt}], \quad A_{\text{out}}[x, y] = A_\chi[x, y]$ for $(x, y) \neq (0, 0)$

Injects 64-bit LFSR round constant into origin lane (prevents round symmetry)

1600

next_state [1599:0]

(Round Feedback)

RC[round_cnt]

64

Control Unit & Sequencer

clock, reset, start

Active-low synchronous reset; single-cycle start pulse

Main Control FSM (3 States)

S_IDLE (2'b00)

start $= 1$

round_en

S_RUN (2'b01)

1 cycle

$r = 23$

S_DONE (2'b10)

count_en

5-bit Round Counter (round_cnt)

Range: $0 \leq$ round_cnt ≤ 23

Clears in S_IDLE; increments each cycle in S_RUN

5

Round Constant ROM (RC[r])

24 precomputed 64-bit constants

$RC[0] = 0\text{x}0001 \dots RC[23] = 0\text{x}8088$

Addressed by round_cnt[4 : 0]

state $=$ S_DONE

ready

Single-cycle active-high completion strobe